

Ishaan Myrie

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EDUCATION

Rutgers University—New Brunswick

B.S. Computer Science, B.A. Mathematics

New Brunswick, NJ

Expected Graduation: May 2029

Relevant Coursework: Machine Learning, Computer Architecture, Stochastic Processes, Linear Optimization, Mathematical Theory of Probability

EXPERIENCE

Rutgers Algorithmic Robotics & Control Lab

Undergraduate Researcher, Applied ML and Robotics

Mar. 2026 – Present

New Brunswick, NJ

- Raised robot task success from 41% to 60% by building a PyTorch pipeline that converted 30K in-house video clips into 3D training data across GPUs with crash recovery.
- Validated 80K data files prior to GPU training by writing a pytest suite integrated into CI.
- Enabled new AI models to be swapped into the training stack without rewriting code by designing a PyTorch encoder that reads a model's dimensions at load time.
- Cut API spend 17x while outperforming every prior method by building the prompting layer for an LLM robot-design tool in Python, targeting the Anthropic Claude API and self-hosted models served with vLLM.

PROJECTS

ResearchTrail | *TypeScript, React, Next.js, Python, FastAPI, PostgreSQL, Cloudflare Workers, Google Cloud*

- Developed an interactive research workspace that helps users visually organize nonlinear research workflows and collaborate across multi-user projects using Next.js, TypeScript, React Flow, and FastAPI.
- Cut production API latency by 46% (2990ms to 1621ms) by moving the backend closer to its database.
- Implemented an autosave workflow with batched PostgreSQL updates, preserving user changes across sessions while reducing unnecessary database writes by 20%.
- Secured 23 of 25 API endpoints with JWT authentication, all enforcing per-user ownership checks, and deployed them through Cloudflare Workers and Google Cloud Run.

Model Evaluation Platform | *Python, Next.js, Supabase, OpenAI API*

- Engineered a repeatable evaluation pipeline that executed 300+ automated test cases, validated model outputs, computed performance metrics, and stored results for comparison across versions.
- Implemented CI/CD quality gates that compared model versions across accuracy, latency, and output-validity metrics, blocking deployment for any version with more than a 4% accuracy drop.

Decision-Support Assistant | *React, TypeScript, Node.js, PostgreSQL*

- Integrated four external APIs into a constraint-aware recommendation engine that evaluated calendar conflicts, user preferences, and travel time to rank feasible options.
- Modeled user preferences, constraints, and recommendation history in PostgreSQL, allowing users to revise inputs and resume prior decision workflows.

PUBLICATIONS

Han, Y., Liu, L., et al. (including **Ishaan Myrie**, 5th author). "A4A: Cross-Embodiment Transfer of Action-Oriented 4D Affordances from Human Demonstrations." [Under Review], 2026.

Liu, L., Yi, P., et al. (including **Ishaan Myrie**, 4th author). "DesignAgent: Efficient and Adaptive Search Framework for Robot Morphology Design." [Under Review], 2026.

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, HTML/CSS, C, SQL

Frameworks: React, Next.js, Node.js, FastAPI, Pydantic, PyTorch, pandas, NumPy, OpenCV

Cloud: Google Cloud Run, Google Cloud Storage, Cloudflare Workers

Tools: Git, CI/CD, pytest, Docker, PostgreSQL, Supabase, Linux